

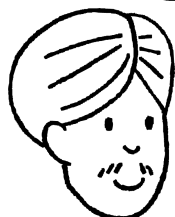


Brackets in equations

Task A: Representations of equations involving brackets

1) Write an equation and draw a bar model for Andeep and Laura's number problems.

I am thinking of a number.
When you multiply it by 5
and add 10 you get 35

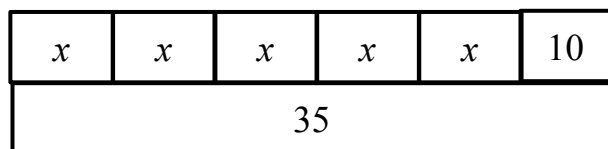


Andeep

$$x \times 5 + 10 = 35$$

$$5x + 10 = 35$$

Simplified form $5(x + 10) = 35$

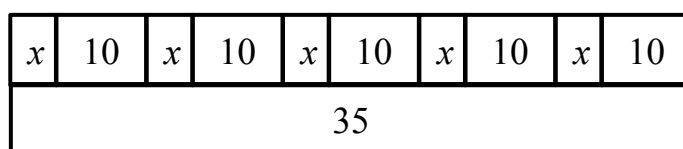


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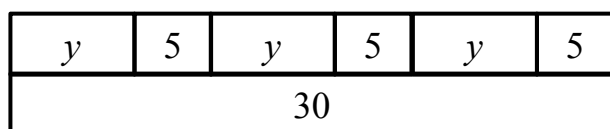


Laura

$$(x + 10) \times 5 = 35$$



2) Write equations for the bar models that Jun and Jacob have drawn. If you can see more than one way to write the equation you should do so.



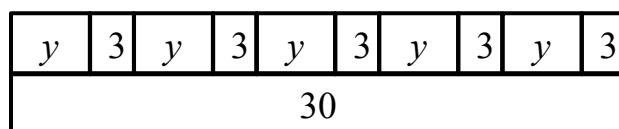
Jun

$$(y + 5) \times 3 = 30$$

$$3(y + 5) = 30$$

$$3 \times y + 3 \times 5 = 30$$

$$3y + 15 = 30$$



Jacob

$$(y + 3) \times 5 = 30$$

$$5(y + 3) = 30$$

$$5 \times y + 5 \times 3 = 30$$

$$5y + 15 = 30$$



Task B: Effectively using brackets in equations

1) Spot the errors in these manipulations of equations.

Once spotted, write a sentence to explain each error.

a) $y + 3 = 4y + 6$

$(4)y + 3 = 4y + 6$

$4y + 12 = 4y + 6$

The multiplication only occurred on one side of the equation.

b) $y + 3 = 4y - 6$

$2(y + 3) = 2(4y - 6)$

$2y + (3) = 8y - (6)$

All terms inside the bracket need to be multiplied.

c) $y + 3 = 4y + 6$

$(4)y + 3 = (2)(4y + 6)$

$4y + 12 = 8y + 12$

Both sides need to be multiplied by the same term.

2) Fill in the missing parts of these manipulations of equations.

a) $2e + 3 = \frac{1}{4}e$

$(4)(2e + 3) = (4)\left(\frac{1}{4}e\right)$

$8e + 12 = e$

b) $\frac{1}{2}d + 4 = 3d - 1$

$(2)\left(\frac{1}{2}d + 4\right) = (2)(3d - 1)$

$d + 8 = 6d - 2$

c) $\frac{7}{f} = 18$

$(f)\left(\frac{7}{f}\right) = (f)(18)$

$7 = 18f$

